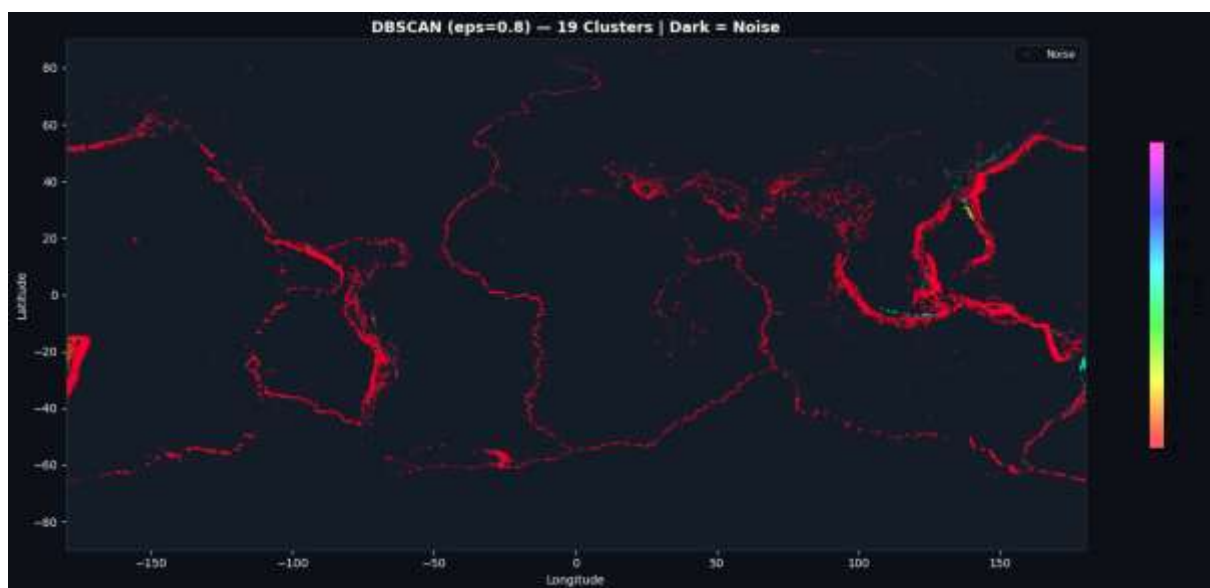
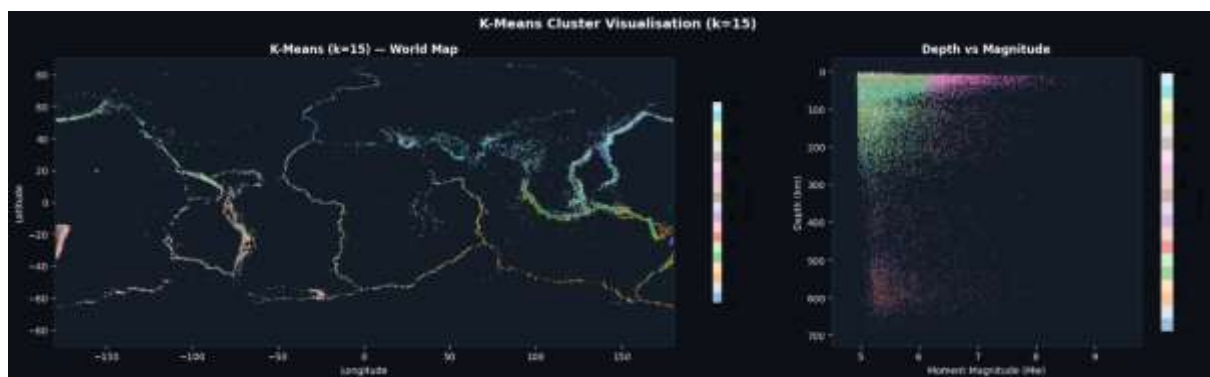
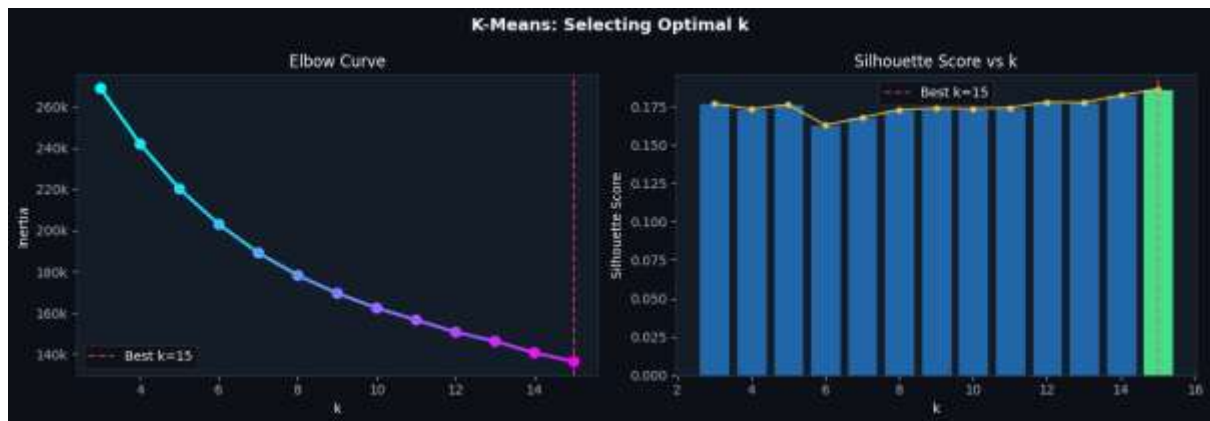
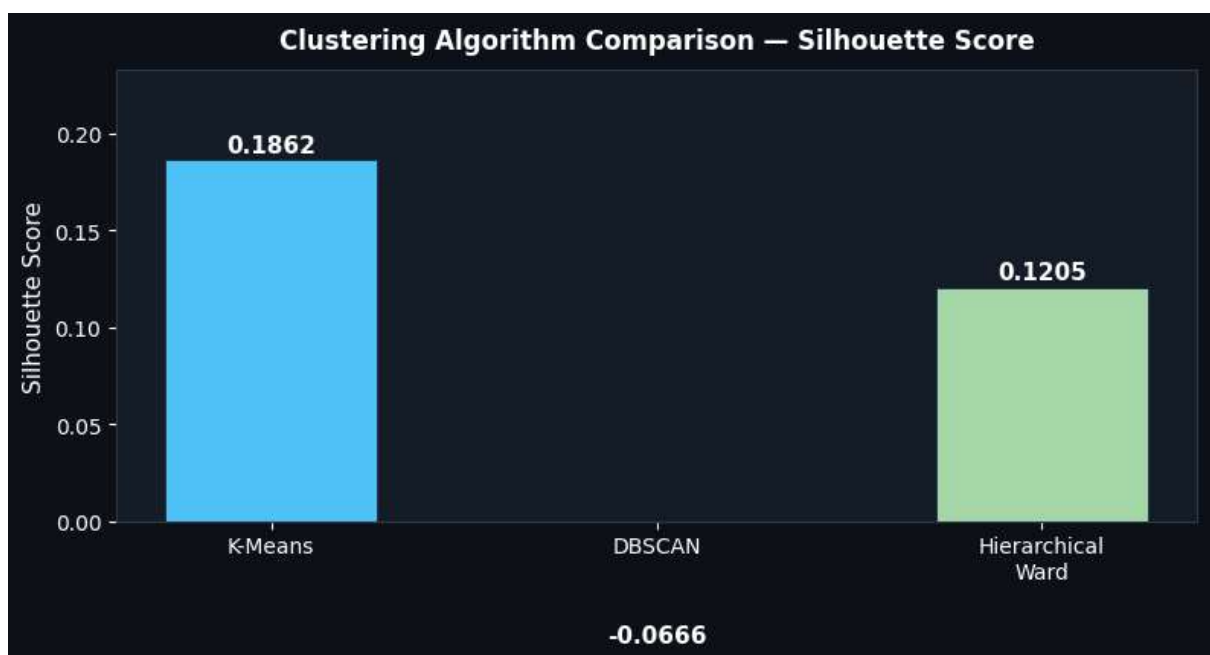
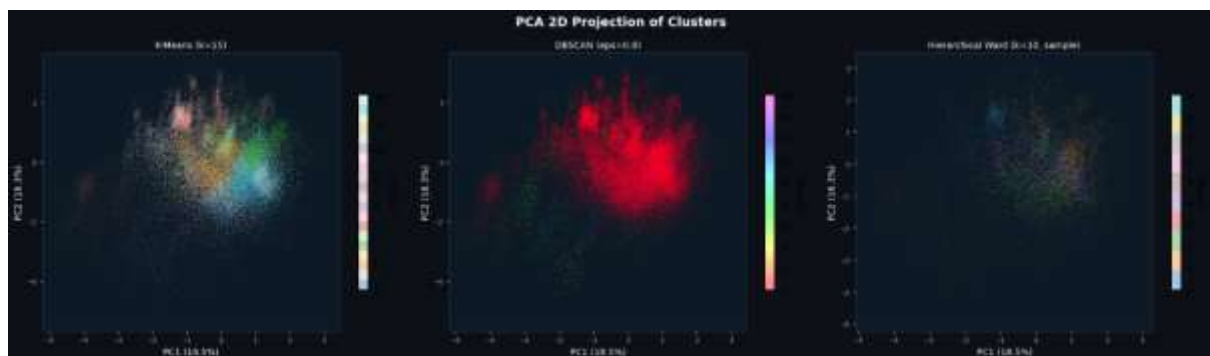
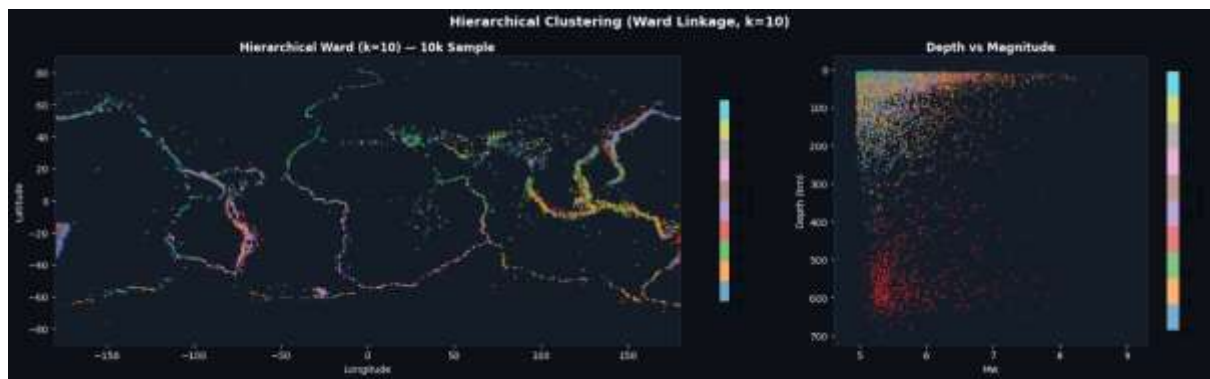
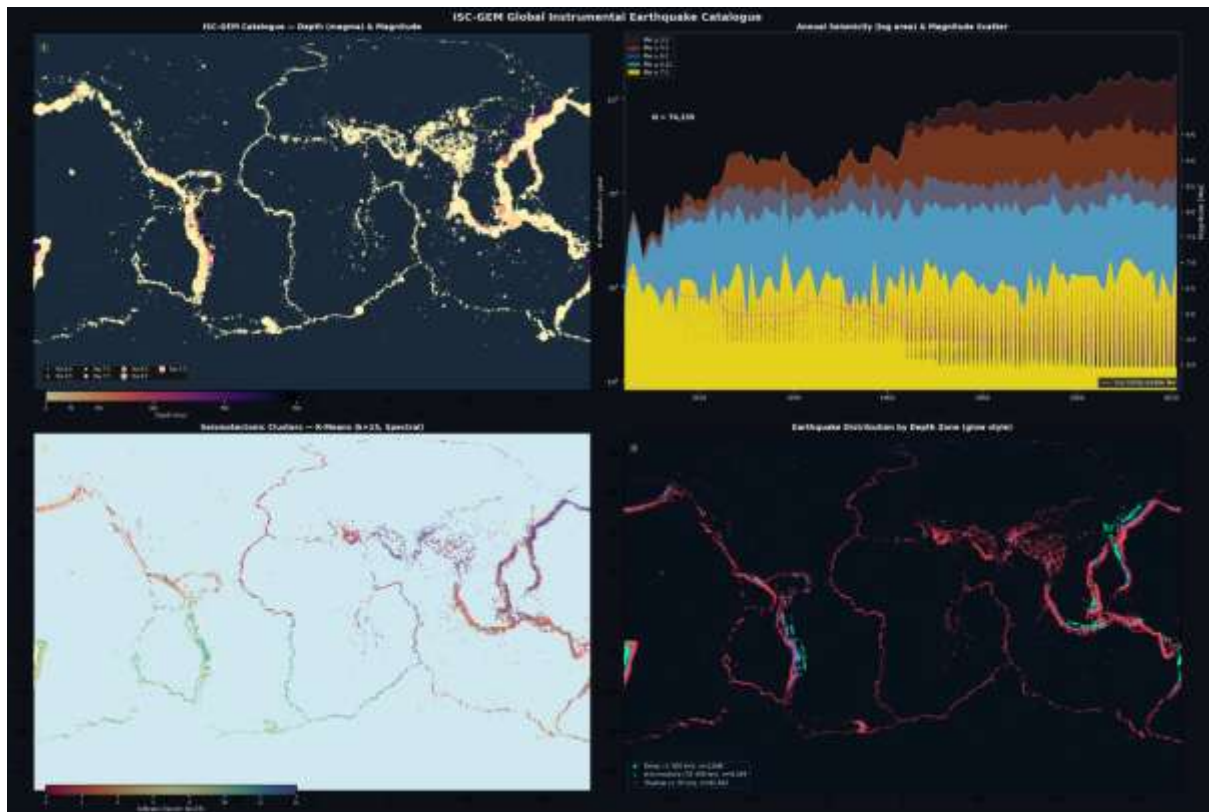








1. Introduction

This project analyses the ISC-GEM Global Instrumental Earthquake Catalogue to understand patterns in earthquake occurrences across the world. The main objective is to study the spatial distribution, depth variation, and magnitude of earthquakes, and to identify groups of similar seismic events using clustering techniques.

The dataset contains key features such as latitude, longitude, depth, magnitude (Mw), and fault type. Using visualization and machine learning methods, the project aims to uncover meaningful geological patterns.

2. Data Description

The dataset includes global earthquake records with the following important attributes:

- **Latitude & Longitude** – represent the geographical location of earthquakes
- **Depth** – indicates how deep the earthquake occurred beneath the Earth's surface
- **Magnitude (Mw)** – measures the strength of the earthquake
- **Fault Type** – describes the nature of tectonic movement (e.g., strike-slip, normal, thrust)

These features are essential for understanding both the physical characteristics and regional patterns of seismic activity.

3. Visualization and Plot Analysis

3.1 Global Earthquake Distribution Plot

The scatter plot of latitude vs longitude shows how earthquakes are distributed across the globe. Most earthquakes are concentrated along tectonic plate boundaries such as the Pacific Ring of Fire, mid-ocean ridges, and subduction zones.

This plot clearly highlights that earthquakes are not randomly distributed but follow geological structures.

3.2 Depth Distribution Plot

The depth plot shows variation in earthquake depth across regions. Most earthquakes occur at shallow depths (0–70 km), but some clusters contain very deep earthquakes (greater than 500 km).

These deep earthquakes are usually associated with subduction zones where one tectonic plate is pushed beneath another.

3.3 Magnitude Distribution Plot

The magnitude histogram indicates that:

- Most earthquakes fall in the moderate range (around Mw 5–6)
- Very high magnitude earthquakes (above Mw 8) are rare

This reflects the natural distribution where smaller earthquakes occur frequently, while large ones are less common but more impactful.

3.4 Fault Type Distribution

The analysis of fault types shows the dominance of:

- **Thrust/Reverse faults** – common in subduction zones
- **Strike-slip faults** – common along transform boundaries
- **Normal faults** – associated with divergent boundaries

This helps relate earthquake behaviour to tectonic processes.

4. Clustering Analysis

Clustering techniques were applied to group earthquakes with similar characteristics. The models used include:

- K-Means Clustering
- DBSCAN
- Hierarchical Clustering

The clustering is based on features such as location, depth, and magnitude.

5. Results and Observations

The clustering results reveal several important insights:

- A total of **15 clusters** were identified, each representing a distinct seismic region or pattern.
- Some clusters correspond to **shallow earthquakes** (depth ~30–50 km), while others represent **deep-focus earthquakes** (depth >500 km).

- The cluster with the highest magnitude recorded **Mw 9.55**, indicating extremely powerful earthquakes in specific regions.
- Deep earthquake clusters (e.g., clusters with depth around 500 km) are fewer but show higher average magnitudes compared to shallow ones

7. Conclusion

This analysis demonstrates that earthquakes are strongly influenced by tectonic structures and are not randomly distributed. Visualization clearly shows concentration along plate boundaries, while clustering helps identify distinct seismic zones with unique characteristics.

The results highlight the importance of depth, magnitude, and fault type in understanding earthquake behaviour. Clustering provides a powerful way to group similar events and gain deeper insights into global seismic activity.